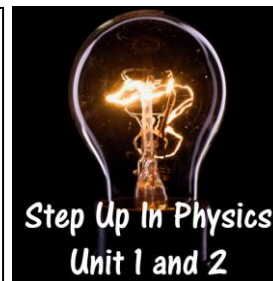


Specific Heat Capacity

Problems Worksheet

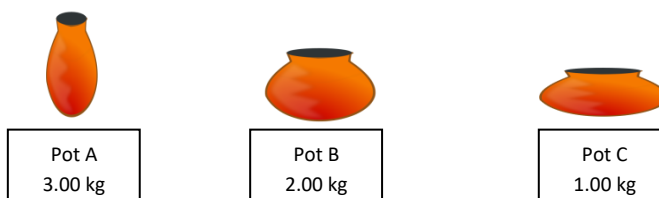


1. Copper has a specific heat capacity of $3.80 \times 10^2 \text{ J kg}^{-1} \text{ K}^{-1}$. What temperature change would 300 g of copper experience when heated with 560 J of thermal energy?

2. The specific heat capacity of a material is the internal energy change when 1.00 kg of that material changes by 1.00 K.
 - a. Calculate the amount of energy needed to increase 0.500 L of water from 25.0°C to 35.0°C .

 - b. Without using calculations but instead referring to the definition of specific heat capacity, describe how this amount of energy would compare to the energy the water would release when cooling from 10.0°C to 0.00°C .

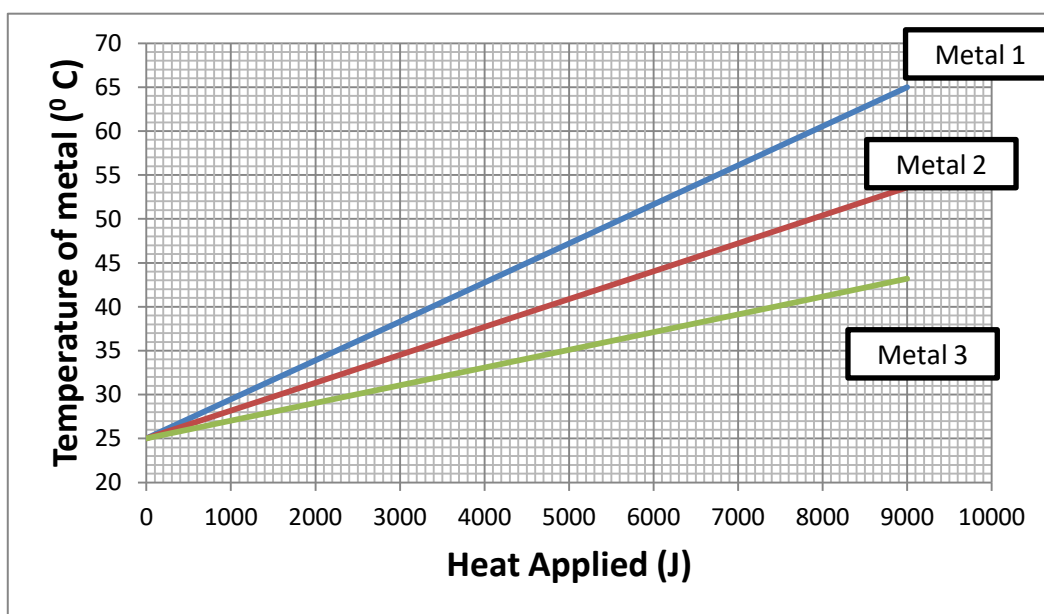
3. The three terracotta pots below differ in shape and mass but have the same surface area.



- a. Which pot (A, B or C) would have the highest specific heat capacity? Justify your choice.

- b. The three pots were all made in the same hot kiln operating at 500°C . Which pot would have cooled down to room temperature first when all were removed from the kiln? Justify your choice.

4. Three different metals of equivalent mass were heated from room temperature and had their temperature recorded as a function of heat applied.



- a. Which metal (1, 2 or 3) has the higher specific heat capacity? Justify your choice.
- b. Metal 3 has a specific heat capacity of $330 \text{ J kg}^{-1} \text{ K}^{-1}$. Use this information and the graph of the heating curve of metal 3 to find the mass of the metals.
- c. Another sample identical to metal 1 is heated from an initial temperature of 35°C . The heat is applied twice as fast. Add a new line to represent the temperature against heat applied heat for this sample.

5. A pizza removed for a few minutes from a hot oven is often cool enough to touch on its crust while the cheese feels much hotter. Explain this observation.
6. A solar hot water system absorbs $8.78 \times 10^6 \text{ J}$ from the Sun which is used to increase the water temperature by from 22.0°C to 34.0°C
- What mass of water was heated by the Sun?
 - Half of the water is used for washing and showering in the evening. Overnight the remaining water fell by 18°C . If the same amount of solar energy reached the hot water system the next day, what is the maximum temperature this water will reach?

7. A cooling system in a car consists of a coolant passed through a closed pipe system to absorb heat energy from the motor and a radiator to transfer that heat to the surrounding atmosphere. A company was testing three new coolants for their effectiveness.

Coolant	Melting Temperature	Boiling Temperature	Specific Heat Capacity ($\text{Jkg}^{-1}\text{K}^{-1}$)
A	-40°C	95°C	4800
B	-30°C	20°C	5000
C	0°C	100°C	3900

Which coolant (A, B or C) would be the best choice for the coolant in a car? Justify your choice by giving reasons why the other two are not suitable choices.

8. Water has a specific heat capacity of $4180 \text{ Jkg}^{-1}\text{K}^{-1}$ while soil and rock has an average specific heat capacity of $850 \text{ Jkg}^{-1}\text{K}^{-1}$.
- Which terrain (water or ground) would cool down faster once the Sun sets in summer?
 - Despite the inner regions of Western Australia being arid desert with higher maximum temperatures in the day, these regions also have lower minimum temperatures on winter nights compared to coastal regions. Explain this observation.

9. A sample of copper and a sample of aluminium of equal mass were heated with the same quantity of heat energy. Aluminium has a specific heat capacity of $900 \text{ J kg}^{-1} \text{ K}^{-1}$. The copper increased in temperature by 5.00°C and has a specific heat capacity of $385 \text{ J kg}^{-1} \text{ K}^{-1}$. Calculate the change in temperature of the aluminium.

10. Ethylene glycol has half the specific heat capacity of water. A sample of ethylene glycol was heated on an element that was set to 80% the power that was used to heat a 1.00 L sample of water. What is the mass of ethylene glycol if its temperature increased at the same rate as the water?

